Original Article

Cardiopulmonary Exercise Testing and Prognosis in Hypertrophic Cardiomyopathy

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Background—Exercise testing is performed in patients with hypertrophic cardiomyopathy to evaluate blood pressure response, a risk factor for sudden cardiac death. The prognostic role of exercise gas exchange variables is unknown.

Methods and Results—Between 1998 and 2010, 1898 patients (age 47±15 years, range 16–86 years; 67% male) with hypertrophic cardiomyopathy underwent cardiopulmonary exercise testing. A total of 178 (9.4%) patients reached the primary end point of all-cause mortality or heart transplant (death/transplant) during a median follow-up of 5.6 years (interquartile range 2.6–8.9), giving an annual event rate of 1.6% per person year. Peak oxygen consumption (adjusted hazard ratio [HR] 0.82, 95% confidence interval [CI] 0.77–0.88, $P<0.001$), ventilatory efficiency (adjusted HR 1.10, 95% CI 1.00–1.22, $P=0.049$), and ventilatory anaerobic threshold (adjusted HR 0.82, 95% CI 0.70–0.96, $P=0.016$) were predictors of the primary outcome after correction for age, sex, left atrial size, nonsustained ventricular tachycardia, and ejection fraction. The overall adjusted death/transplant estimates for patients in the lowest quartile with peak oxygen consumption ≤15.3 mL/kg/min were 14% at 5 years and 31% at 10 years. Peak oxygen consumption (HR 0.81, 95% CI 0.77–0.86, $P<0.01$) and ventilation to carbon dioxide production (HR 1.10, 95% CI 1.08–1.13, $P<0.001$) were predictors of death because of heart failure or transplantation but not sudden cardiac death or implantable cardioverter defibrillator shocks.

Conclusions—Cardiopulmonary exercise testing provides prognostic information in patients with hypertrophic cardiomyopathy. Submaximal exercise parameters, such as ventilatory efficiency and anaerobic threshold, measured alone or in combination with peak oxygen consumption, predict death from heart failure. **(*Circ Heart Fail*. 2015;8:1022-1031. DOI: 10.1161/CIRCHEARTFAILURE.114.002248.)**

Key Words: exercise testing ■ heart failure ■ hypertrophic cardiomyopathy ■ prognosis ■ sudden cardiac death

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Hypertrophic cardiomyopathy (HCM) is an inherited heart condition characterized by a complex and variable clinical phenotype. The disease usually progresses slowly with many years of stable or minor symptoms, and around a quarter of those affected have normal life expectancy.[1,2] Sudden cardiac death (SCD) can occur throughout life but recent cohort studies show that heart failure (HF) and stroke-related deaths are increasingly important causes of morbidity and mortality in the long term.[3,4]

Clinical Perspective on p 1031

Cardiopulmonary exercise testing (CPX) provides a noninvasive method for assessing the cardiovascular, pulmonary, and skeletal muscle components of exercise performance.[5] In cardiovascular disease, CPX has many applications, including the evaluation of patients with systolic and diastolic HF, pulmonary hypertension, dilated cardiomyopathy, and congenital heart disease.[6–11] Patients with HCM routinely undergo exercise testing (usually without gas exchange measurement) to assess blood pressure response, a marker of sudden death risk in young adults,[12,13] and to identify exercise-induced outflow tract obstruction and ventricular arrhythmia.[14,15] CPX has been used to differentiate HCM from physiological ventricular hypertrophy,[16] but its role in predicting clinical outcome remains undetermined.[17,18]

The majority of studies examining the prognostic value of CPX have used peak oxygen consumption (peak $\dot{V}O_2$) as the primary measure of exercise performance. However, in a disease population, peak $\dot{V}O_2$ may be influenced by physical conditioning and motivation. For this reason, measurement of submaximal exercise parameters that are less dependent on maximal effort, such as ventilatory efficiency (minute ventilation to carbon dioxide production [$\dot{V}_E\dot{V}CO_2$]) and ventilatory anaerobic threshold (VAT), have been advocated.[19–21]

In this study, we present a single-center experience of CPX in a large unselected cohort of patients with HCM in which we sought to determine the relation between maximal and submaximal gas exchange parameters and long-term survival.

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Methods

Study Design

This is a single-center observational cohort study of consecutive patients with HCM who underwent CPX between March 1998 and December 2010. CPX data were examined retrospectively. The study conforms to the principles of the Helsinki declaration. Collection of data for prognostic modeling without written consent using routinely collected clinical data was approved by the local research ethics committee (REC 14/WS/0102). Only patients aged ≥16 years were included. A diagnosis of HCM was based on echocardiographic identification of left ventricular (LV) hypertrophy (maximum LV wall thickness ≥15 mm) unexplained solely by loading conditions or in accordance with the criteria for the diagnosis of familial disease in patients with at least 1 first-degree relative with an unequivocal diagnosis.[22,23] Patients with Friedreich's ataxia, Noonan syndrome, Anderson–Fabry disease, and other metabolic or multisystem disorders associated with cardiac hypertrophy were excluded. Patients unable or unwilling to undergo CPX testing were excluded from the analysis.

Clinical Assessment

A cardiologist evaluated all patients. Symptoms were documented, including chest pain, palpitations, and New York Heart Association functional class. Clinical risk stratification was performed in accordance with previously published methods.[24] Medication at the time of CPX was recorded. A 12-lead ECG was recorded at 25 mm/s and 10 mm/mV before CPX. Transthoracic 2D and Doppler echocardiography was performed in all patients. Maximum LV wall thickness was defined as the greatest thickness in any single LV segment. LV end-diastolic and end-systolic diameter were measured using M-mode and 2D images obtained from parasternal long-axis views, and fractional shortening and Teicholz ejection fraction (EF) were calculated.[25] LV outflow tract (LVOT) gradient was calculated from continuous-wave Doppler using the simplified Bernoulli equation. LVOT obstruction (LVOTO) was defined as a peak gradient ≥30 mm Hg at rest.[26] Interventions to treat LVOTO, including septal myectomy, mitral valve replacement, alcohol septal ablation, and dual chamber pacemaker implantation, were recorded but not used as censoring events in the survival analysis.

CPX Testing

Exercise testing was performed in a dedicated air-conditioned exercise laboratory at St George's Hospital (pre-2003) or The Heart Hospital (post-2003) supervised by an exercise physiologist (B. Mist). Height and weight were recorded and body mass index calculated. All patients performed upright bicycle exercise, after a period of resting and unloaded pedalling, to maximum tolerance with the use of a progressively increasing work rate of 10 to 30 watts/min selected by the physiologist according to the patients' age and reported exercise capacity.[27] Subjects cycled at a constant rate of 60 to 65 revolutions per minute (rpm) until exhaustion. Respiratory gas exchange analysis was performed using SensorMedics (VIASYS, Yorba Linda, CA) metabolic cart and Vmax Encore 229 software. Peak $\dot{V}O_2$ was defined as the mean of the highest 2 consecutive values of 20-second averages of $\dot{V}O_2$. Percentage predicted $\dot{V}O_2$ was calculated using the regression equations published by Wasserman et al.[28] The peak respiratory exchange ratio was defined as the ratio between $\dot{V}CO_2$ and $\dot{V}O_2$ at peak exercise. The VAT was measured by the V-slope method supplemented by additional parameters, including change in respiratory exchange ratio, end tidal P_{O_2} increase, and $V_E/\dot{V}CO_2$ increase.[29,30] The $V_E/\dot{V}CO_2$ slope was obtained by linear regression analysis of the data acquired throughout the entire period of exercise.[8] Heart rate reserve was calculated as the difference between peak and resting heart rates. Patients with a cardiac pacemaker, sustained atrial tachyarrhythmia, or second- or third-degree AV block during exercise testing were excluded from the heart rate reserve analyses. A normal blood pressure response was defined as an increase in the systolic pressure of >20 mm Hg from rest to peak exercise in the absence of a fall of >20 mm Hg from peak pressure.[12,13] Rate pressure product was calculated as peak heart rate multiplied by peak systolic blood pressure.

Study End Points

The primary end point was all-cause mortality or cardiac transplantation. This was chosen to eliminate bias arising from inadvertent misclassification of the cause of death. The secondary end points were (1) combined cardiovascular deaths (HF, cardiac transplant, stroke, SCD, and other cardiac deaths, (2) death caused by HF or cardiac transplantation, (3) SCD, and (4) SCD, aborted SCD, and appropriate implantable cardioverter defibrillator shocks. Death related to HF was defined as death preceded by signs and symptoms of HF of >1 hour duration or cardiogenic shock. SCD was defined as witnessed SCD within 1 hour of new symptoms and nocturnal deaths with no antecedent history of worsening symptoms. Aborted SCD describes an event where an intervention (eg, defibrillation) or spontaneous reversion restores circulation. Implantable cardioverter defibrillator shocks were considered appropriate if the treated tachyarrhythmia was ventricular in origin. Definitions are consistent previous studies.[31]

Follow Up

Patients were followed in the outpatient department at The Heart Hospital. Survival status and time to death was assessed through clinic attendance and the National Health Service computer system. The cause of death was ascertained from review of the clinical notes and postmortem report, if performed. The last date of follow up was June 27, 2013.

Statistical Analysis

All statistical analyses were performed using STATA (Version 13). Variables were summarized using mean±standard deviation, median and interquartile range, or counts and percentages, as appropriate. The follow-up time for each patient was calculated from the date of their first CPX test to the date of their most recent National Health Service encounter or censoring event up to June 27, 2013. The annual event rate was calculated by dividing the number of patients reaching an end point by the total follow-up period and expressed as a percentage. The cumulative probability for the occurrence of an end point was estimated using the Kaplan–Meier method.

Univariate analyses using Cox proportional hazard model were performed to test the association between the predictor variables and each of the time to event outcomes (primary and secondary).[32] A minimum of 10 events per parameter estimated by the model is required to avoid overfitting. For the primary outcome (death or heart transplant), a total number of 178 events were observed which would allow the estimation of 18 regression coefficients. The key predictor variables were $\dot{V}O_2$, $V_E/\dot{V}CO_2$, and VAT. In addition to demographic variables which include age at time of CPX and sex, the following variables were considered to be important based on clinical experience and previous published literature: New York Heart Association Class, LVOT gradient, EF, left atrium size, nonsustained ventricular tachycardia, family history of SCD, previous syncope, and abnormal blood pressure response. The multivariable adjusted Cox model included all exercise parameters, age, and sex. For the primary outcome, a 20% significance level was used in the backward elimination procedure to select additional predictors for the multivariable model. A square term was added to continuous predictors with nonlinear functional association.

The proportional hazards assumption required by the Cox model was investigated using Schoenfeld residual.[33] The C-index (C-Uno) and D-statistic were used to describe the models' discriminations.[34,35] A value of 0.5 for C-index indicates no discrimination, and a value equal to 1 indicates perfect discrimination. The D-statistic quantifies the observed separation between subjects with low and high predicted risks as predicted by the model and can be interpreted as the log hazard ratio for having an event between the low and high risk groups of patients. A model with no discriminatory ability will produce a value of 0 for D-statistic, with increasing values indicating greater separation.

The probability of all-cause mortality or transplant at a particular time point for an individual patient was estimated based on the average survival probability at that time and the sum of the product of the predictors and their coefficient in the multivariable model (linear predictor) using the following equation:

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Table 1. Comparison of Demographic and Clinical Characteristics of Study Population by Quartiles of Peak $\dot{V}O_2$

	All Patients (n=1898)	Quartile 1 (n=479)	Quartile 2 (n=482)	Quartile 3 (n=466)	Quartile 4 (n=471)	*P* Value*
Peak $\dot{V}O_2$, mL/kg/min		≤15.3	15.4–20.8	20.9–27.2	≥27.3	
Males	1278 (67%)	194 (41%)	296 (61%)	353 (76%)	435 (92%)	<0.001
Age, y	46±15	56±14	49±14	45±13	37±12	<0.001
Ethnicity						<0.001
a. White Caucasian	1442 (76%)	339 (71%)	371 (77%)	361 (77%)	371 (79%)	
b. Black Afro-Caribbean	75 (4%)	31 (6%)	25 (5%)	12 (3%)	7 (1%)	
c. Asian	143 (8%)	54 (11%)	32 (7%)	37 (8%)	20 (4%)	
d. Mixed	29 (2%)	6 (1%)	12 (2%)	10 (2%)	1 (0.2%)	
e. Not stated or unknown	209 (11%)	49 (10%)	42 (9%)	46 (10%)	72 (15%)	
Height, m	1.71±0.09	1.66±0.09	1.70±0.09	1.73±0.08	1.77±0.08	<0.001
Weight, kg	81±16	80±17	82±18	81±17	80±14	0.305
Body mass index, kg/m^2	27±5	29±5	28±5	27±4	25±4	<0.001
Clinical characteristics						
Familial history of HCM	778 (42%)	153 (33%)	188 (40%)	199 (43%)	238 (51%)	<0.001
Family history of SCD because of HCM	321 (17%)	66 (14%)	70 (15%)	86 (18%)	99 (21%)	0.025
Symptoms						
NYHA class III or IV	145 (8%)	106 (23%)	23 (5%)	12 (3%)	4 (1%)	<0.001
Chest pain	757 (41%)	249 (53%)	215 (45%)	172 (37%)	121 (26%)	<0.001
Palpitations	556 (30%)	187 (40%)	163 (34%)	126 (27%)	80 (17%)	<0.001
Syncope	216 (12%)	70 (15%)	65 (14%)	37 (8%)	44 (10%)	0.002
Medication						
Calcium channel or β-adrenoreceptor blocker	970 (51%)	316 (67%)	300 (63%)	231 (50%)	123 (26%)	<0.001
Arrhythmia						0.267
Nonsustained VT	531 (28%)	131 (27%)	151 (31%)	136 (30%)	113 (24%)	
History of VT or VF	11 (1%)	3 (1%)	2 (0.5%)	2 (0.5%)	4 (1%)	
Procedure at baseline						
Septal myectomy	31 (2%)	14 (3%)	9 (2%)	5 (1%)	3 (1%)	0.030
Alcohol septal ablation	17 (1%)	8 (2%)	4 (1%)	3 (1%)	2 (0.5%)	0.189
Pacemaker	88 (5%)	49 (10%)	19 (4%)	10 (2%)	10 (2%)	<0.001
Defibrillator	63 (3%)	24 (5%)	24 (5%)	8 (2%)	7 (4%)	0.001
Procedure during FU						
Septal myectomy	182 (10%)	65 (14%)	66 (14%)	33 (7%)	18 (4%)	<0.001
Alcohol septal ablation†	43 (2%)	17 (4%)	10 (2%)	11 (2%)	5 (1%)	0.080
Pacemaker	109 (6%)	44 (9%)	39 (8%)	19 (4%)	7 (1%)	<0.001
Defibrillator	338 (18%)	64 (13%)	95 (20%)	94 (20%)	85 (18%)	0.024
ECG						<0.001
a. Sinus rhythm	1545 (94%)	335 (88%)	393 (92%)	401 (96%)	416 (98%)	
b. Atrial fibrillation	73 (4%)	30 (8%)	26 (6%)	13 (3%)	4 (1%)	
c. Paced rhythm	34 (2%)	17 (4%)	10 (2%)	4 (1%)	3 (1%)	
Echocardiogram						
Max wall thickness, mm	19±5	18±5	19±5	18±5	18±5	0.008
LA diameter, mm	44±8	46±8	46±8	44±7	42±7	<0.001
LVED, mm	46±6	45±7	45±6	45±6	46±6	<0.001
LVES, cm	27±6	28±7	27±7	27±6	27±6	0.548
Ejection fraction, %	65±11	63±12	64±11	65±10	66±10	0.011
Fractional shortening, %	40±9	39±10	40±9	41±8	42±9	<0.001
Mod or severe MR	221 (12%)	90 (19%)	67 (14%)	41 (9%)	23 (5%)	<0.001
LVOT obstruction ≥30 mm Hg	568 (31%)	196 (43%)	159 (35%)	125 (29%)	88 (19%)	<0.001
LVH ≥30 mm	70 (4%)	12 (3%)	27 (6%)	17 (4%)	14 (3%)	0.056

Characteristics are reported as mean±SD or n (%). HCM indicates hypertrophic cardiomyopathy; LA, left atrium; LVED, left ventricular end-diastolic diameter; LVES, left ventricular end-systolic diameter; LVH, left ventricular hypertrophy; LVOT, left ventricular outflow tract obstruction; MR, mitral regurgitation; NYHA, New York Heart Association; SCD, sudden cardiac death; VT, ventricular tachycardia; VF, ventricular fibrillation; and $\dot{V}O_2$, oxygen consumption.

**P* values were obtained from Chi-square test, Fisher exact test, or ANOVA.

†Four patients underwent both alcohol septal ablation and myectomy during follow-up; they are not included in the table.

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Table 2. Comparison of Exercise Data by Primary and Secondary Outcomes

	Population	Primary Outcome		Secondary Outcomes			
	All patients (n=1898)	Survivors (n=1720)	Death/Transplant (n=178)	CV Death (n=117)	SCD (n=40)	HF/Transplant (n=53)	SCD Plus ICD (n=74)
Peak $\dot{V}O_2$, L/min	1.77±0.80	1.81±0.79	1.35±0.72*	1.45±0.76	1.99±0.88	1.07±0.44	1.88±0.81
Peak $\dot{V}O_2$, mL/kg/min	22.0±9.1	22.4±9.0	17.5±8.9*	18.7±9.7	25.6±11.7	14.1±5.2	24.0±10.3
Peak $\dot{V}O_2$, % predicted	67±21	68±20	58±23*	59±22	71±22	46±17	67±23
Heart rate at rest, bpm	71±15	71±15	71±16	72±17	73±13	71±17	72±16
Heart rate at peak, bpm	138±30	140±29	125±33*	131±33	148±29	121±29	146±30
Heart rate reserve, bpm	68±27	69±27	54±30*	59±32	75±29	49±29	74±27
Systolic BP rest, mm Hg	126±21	126±21	126±23	123±22	123±21	118±17	120±20
Systolic BP peak, mm Hg	171±35	172±34	158±34*	155±36	162±33	145±36	158±34
Abnormal BP response	406 (21%)	344 (20%)	62 (35%)*	42 (36%)	11 (28%)	27 (51%)	22 (30%)
RPP ($\times 10^{-2}$), mm Hg/min	239±75	243±74	201±69*	207±73	240±70	176±63	231±70
Respiratory exchange ratio	1.10±0.11	1.10±0.11	1.09±0.13	1.10±0.13	1.11±0.10	1.11±0.14	1.13±0.11
$\dot{V}_E\dot{V}CO_2$ slope	32.6±7.3	32.1±6.8	37.8±10.0*	37.2±10.0	32.9±5.9	41.0±11.9	32.6±7.0
Patients reaching AT	1789 (94%)	1637 (95%)	152 (85%)*	97 (83%)	36 (90%)	41 (77%)	69 (93%)
$\dot{V}O_2$ at AT, mL/kg/min	11.7±4.2	11.8±4.2	10.4±4.3*	11.1±4.5	13.8±5.1	9.1±2.6	13.1±4.4
Peak workload, watts	129±67	133±67	94±60*	105±64	148±74	74±37	142±68
Exercise duration, min	8.2±2.4	8.3±2.4	6.9±2.6*	7.2±2.7	8.4±2.4	6.4±2.3	8.3±2.4

AT indicates anaerobic threshold; BP, blood pressure; RPP, rate pressure product; and $\dot{V}O_2$, oxygen consumption.
*P values are <0.05 obtained for the primary outcome using Chi-square test, Fisher exact, or independent t test.

$$\hat{P}_{\text{event at time } t} = 1 - S_0(t)^{\exp(\text{linear predictor})}$$

where $S_0(t)$ is the average probability at time t (eg, 5 years). Graphical comparisons of the observed and predicted all-cause mortality or transplant at 5 years by quartiles $\dot{V}O_2$ at 5 years and 10 years were performed.

Results

Study Population

One thousand nine hundred and fifty-one patients were eligible for the study. Of these, 53 (3%) patients were excluded because they were unable or unwilling to undergo exercise testing. Baseline characteristics of the remaining 1898 patients are shown in Table 1.

Follow Up

Data on 1898 patients were available. However, 72 patients did not contribute to the survival analysis because they were seen only once. A total of 178 (9.4%) patients reached the primary end point of all-cause mortality or heart transplant (mortality/transplant) during a median (interquartile range) follow up of 5.6 years (interquartile range 2.6–8.9). Patients were followed for median 5.6 years (interquartile range 2.6–8.9; range 0.003–14.90), during which time 178 death/transplant events were observed. The overall death/transplant rate was 1.6% per person-year. There were 117 cardiovascular end points, including 40 sudden deaths, 31 deaths caused by HF, 22 cardiac transplants, and 24 others (strokes, myocardial infarction, systemic thromboembolism, and postprocedural deaths). At baseline, 63 (3%) patients had an implantable cardioverter defibrillator, which increased to 401 (21%) by the end of the follow-up period. Two hundred and twenty-nine patients (12%) underwent procedures to reduce LVOTO (182 myectomy, 43 alcohol septal ablations, and 4 both procedures).

Cardiopulmonary Exercise Testing

CPX variables are presented according to the primary and secondary outcomes in Table 2. All peak exercise responses were higher in survivors, including heart rate, blood pressure, peak

Table 3. Estimate of Risk of Death or Transplant Based on a Cox Regression Model (Univariate Analyses, n=1826)

Variable	HR	95% CI	P Value
Exercise parameters*			
Peak $\dot{V}O_2$	0.79	0.74–0.83	<0.001
V_E $\dot{V}CO_2$	1.18	1.08–1.30	<0.001
VAT	0.71	0.62–0.82	<0.001
Demographics			
Age†	1.03	1.02–1.05	<0.001
Female sex	1.61	1.20–2.17	0.002
Clinical Parameters			
NYHA Class III/IV	3.41	2.28–5.10	<0.001
Ejection fraction	0.07	0.02–0.27	<0.001
MWT	0.99	0.96–1.02	0.365
LVOT gradient	1.00	1.00–1.01	0.028
Rate limiting medication	1.21	0.90–1.63	0.197
Left atrial size	1.07	1.05–1.09	<0.001
Nonsustained VT	1.66	1.23–2.23	0.001
Family history of SCD	1.01	0.70–1.47	0.948
Abnormal BPR	2.15	1.58–2.94	<0.001
Syncope	1.37	0.90–2.08	0.140

BP indicates blood pressure; BPR, BP response; CI, confidence interval; HR, hazard ratio; MWT, maximum wall thickness; LVOT, left ventricular outflow tract obstruction; NYHA, New York Heart Association; SCD, sudden cardiac death; VAT, ventilatory anaerobic threshold; $\dot{V}O_2$, oxygen consumption; and VT, ventricular tachycardia.
*Estimates obtained with square term added because of evidence of nonlinearity. Hazard ratios for exercise parameters without square terms added were peak $\dot{V}o_2$ 0.91 (0.89, 0.93), V_E $\dot{V}CO_2$ 1.07 (1.06, 1.90), and VAT 0.87 (0.84, 0.92).
†Log transformation of age provided a better fit (HR 3.63 [95% CI 2.17–5.85], P<0.001).

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Table 4. Adjusted Hazard Ratios (95% CI) for the Primary Outcome (Death or Transplant) and Its Components Based on Cox Regression Model Considering Individual Exercise Parameters and the Combination of All Three Exercise Parameters (peak $\dot{V}O_2$, $\dot{V}_E\dot{V}CO_2$, and VAT)

	Death or Transplant (n=178)		Death (n=156)		Transplant (n=22)	
Exercise Variables	HR (95% CI)	*P* Value	HR (95% CI)	*P* Value	HR (95% CI)	*P* Value
a. Individual parameters*						
Peak $\dot{V}O_2$	0.82 (0.77, 0.88)	<0.001	0.82 (0.77, 0.88)	<0.001	0.87 (0.79, 0.96)	0.004
Goodness-of-fit						
D-statistic	1.63		1.65		2.75	
C-index	0.76		0.76		0.89	
C-Uno	0.78		0.77		0.91	
$\dot{V}_E\dot{V}CO_2$	1.10 (1.00, 1.22)	0.049	1.13 (1.01, 1.26)	0.030	1.07 (1.03, 1.12)	0.002
Goodness-of-fit						
D-statistic	1.65		1.61		3.05	
C-index	0.76		0.76		0.87	
C-Uno	0.77		0.76		0.90	
VAT	0.82 (0.70, 0.96)	0.016	0.79 (0.67, 0.93)	0.005	1.01 (0.85, 1.21)	0.883
Goodness-of-fit						
D-statistic	1.56		1.47		2.41	
C-index	0.73		0.74		0.85	
C-Uno	0.74		0.72		0.88	
b. Combined parameters*						
Peak $\dot{V}O_2$, mL/kg/min	0.85 (0.77, 0.92)	<0.001	0.85 (0.78, 0.93)	0.001	0.82 (0.70, 0.96)	0.013
$\dot{V}_E\dot{V}CO_2$ slope	1.05 (1.02, 1.07)	<0.001	1.05 (1.02, 1.08)	<0.001	1.02 (0.95, 1.09)	0.651
VAT, mL/kg/min	1.07 (0.98, 1.16)	0.115	1.04 (0.96, 1.14)	0.312	1.32 (1.02, 1.70)	0.032
Goodness-of-fit						
D-statistic	1.57		1.56		2.77	
C-index	0.76		0.76		0.88	
C-Uno	0.79		0.78		0.91	
c. Clinical parameters						
Female sex	1.90 (1.40, 2.59)	<0.001	1.59 (1.14, 2.21)	0.006	6.86 (2.60, 18.06)	<0.001
Age	0.89 (0.84, 0.94)	<0.001	0.87 (0.82, 0.92)	<0.001	1.22 (0.91, 1.65)	0.190
Ejection fraction	.00003 (1.67e^{-08}, 0.081)	0.009	0.005 (3.46e^{-07}, 62.06)	0.268	7.59e^{-11} (1.33e^{-17}, 0.0004)	0.003
LA size	1.06 (1.04, 1.08)	<0.001	1.06 (1.04, 1.08)	<0.001	1.09 (1.04, 1.14)	<0.001
NSVT	1.52 (1.12, 2.07)	0.007	1.34 (0.96, 1.87)	0.081	4.15 (1.57, 11.0)	0.004
Goodness-of-fit						
D-statistic	1.51		1.49		2.87	
C-index	0.74		0.73		0.89	
C-Uno	0.74		0.74		0.90	

CI indicates confidence interval; HR, hazard ratio; LA, left atrium; NSVT, nonsustained ventricular tachycardia; VAT, ventilatory anaerobic threshold; and $\dot{V}O_2$, oxygen consumption.

*The model to predict the primary outcome and its components was adjusted for age, sex, ejection fraction, LA size, and NSVT.

$\dot{V}O_2$, VAT, exercise duration, and workload. $\dot{V}_E\dot{V}CO_2$ slope was lower in survivors. Peak $\dot{V}O_2$ values were normally distributed (mean 71%, range 8% to 167% of their predicted value), and 166 patients (10%) had a peak $\dot{V}O_2$ exceeding 100% predicted value for age, sex, and height. Anaerobic threshold was not determined in 109 (6%) patients. Blood pressure response was abnormal in 406 (21%) patients and could not be determined in 22 (1.2%) patients. Twenty-seven patients (1.4%) had nonsustained ventricular tachycardia during exercise. There was a correlation between peak $\dot{V}O_2$ and VAT (r=0.89, P<0.01). There was a negative moderate correlation between peak $\dot{V}O_2$ and $\dot{V}_E\dot{V}CO_2$ (r=–0.49, P<0.01) and between $\dot{V}_E\dot{V}CO_2$ and VAT (r=–0.45, P<0.01).

Clinical and CPX Predictors of Outcome

Results from the univariate Cox model used to examine the relationship between each exercise parameters (peak $\dot{V}O_2$, $\dot{V}_E\dot{V}CO_2$ slope, and VAT) and the prespecified clinical and demographic

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variables as predictors of the relevant outcomes are shown in Table 3. The relationship between all exercise parameters and time to death or transplant was significant (P<0.001). The risk (95% confidence interval [CI]) of death or transplant is reduced by 21% (11% to 26%) for each 1 mL/kg/min increase in peak $\dot{V}O_2$ and by 29% (28% to 38%) for each 1 mL/kg/min increase in VAT. The risk (95% CI) of death or transplant is increased by 18% (8% to 30%) for each 1 U increase in $\dot{V}_E\dot{V}CO_2$.

Other univariable predictors of the primary outcome included age, female sex, New York Heart Association Class III or IV, EF, left atrial size, nonsustained ventricular tachycardia, resting LVOT gradient, and an abnormal blood pressure response (Table 3). There was no statistically significant relationship between death/transplant and maximal LV wall thickness (hazard ratio [HR] 0.99, 95% CI 0.96–1.02, P=0.365), syncope (HR 1.37, 95% CI 0.90–2.08, P=0.14), or family history of SCD (HR 1.01, 95% CI 0.70–1.47, P=0.948). On multivariate Cox analysis, age, sex, left atrial size, EF, and nonsustained ventricular tachycardia were independent predictors of the primary outcome and were used to adjust the final model (Table 4). Survival rates evaluated by a Kaplan–Meier method were reduced in patients with lower peak $\dot{V}O_2$ and VAT values and elevated with $\dot{V}_E\dot{V}CO_2$ slopes (Figure 1).

$\dot{V}O_2$ (adjusted HR 0.82, 95% CI 0.77–0.88, P<0.001), ventilatory efficiency, $\dot{V}_E\dot{V}CO_2$ slope (adjusted HR 1.10, 95% CI 1.00–1.22, P=0.049), and VAT (adjusted HR 0.82, 95% CI 0.70–0.96, P=0.016) were predictors of the primary outcome. The effect of combining all 3 CPX variables was examined with adjustment for the independent clinical predictors (Table 4). Peak $\dot{V}O_2$ (adjusted HR 0.85, 95% CI 0.77–0.92, P<0.001) and $\dot{V}_E\dot{V}CO_2$ slope (adjusted HR 0.85, 95% CI 0.77–0.92, P<0.001) were both independent predictors of the primary outcome (C-index 0.76, D-statistic 1.57).

Secondary Outcomes

Peak $\dot{V}O_2$, VAT, and $\dot{V}_E\dot{V}CO_2$ were predictors of cardiovascular death and death from HF or cardiac transplantation (Table 5). VAT but not peak $\dot{V}O_2$ and $\dot{V}_E\dot{V}CO_2$ slope was a significant predictor of SCD (HR 1.09, 95% CI 1.01–1.16, P<0.017) with higher VAT associated with adverse outcome. Current guidelines recommend that cardiac transplantation be considered in patients with a peak $\dot{V}O_2$ of ≤14 mL/kg/min.[36] Three hundred and sixty-nine patients had a peak $\dot{V}O_2$ of ≤14 mL/kg/min. The annual event rate of transplant or HF in this group was 1.63% (95% CI 1.03–2.24) compared with 0.27% (95% CI 0.16–0.37) in patients with a peak $\dot{V}O_2$ of >14 mL/kg/min. Of the 22 patients undergoing transplantation, 11 patients had a peak $\dot{V}O_2$ ≤14 mL/kg/min at baseline (range 6–24.5 mL/kg/min).

Sub-Analyses

Sub-analyses using the $\dot{V}_E\dot{V}CO_2$ slope up to the ventilatory compensation point instead of calculating the slope from entire exercise period were also performed. No major differences in the predictive value of these 2 methods were found. Similarly no significant differences were found if patients achieving a respiratory exchange ratio <1 were excluded. The 109 patients who did not reach anaerobic threshold had a 3-fold risk of reaching the primary end point (relative risk, 2.81, 95% CI 1.94–4.06, P<0.001). When stratified according to quartiles of $\dot{V}O_2$, patients with LVEF ≥50% and LVEF <50% exhibited a stepwise increase in hazard rates of mortality/transplant, although subjects with LVEF ≥50% had lower overall event rates (Figure 2).

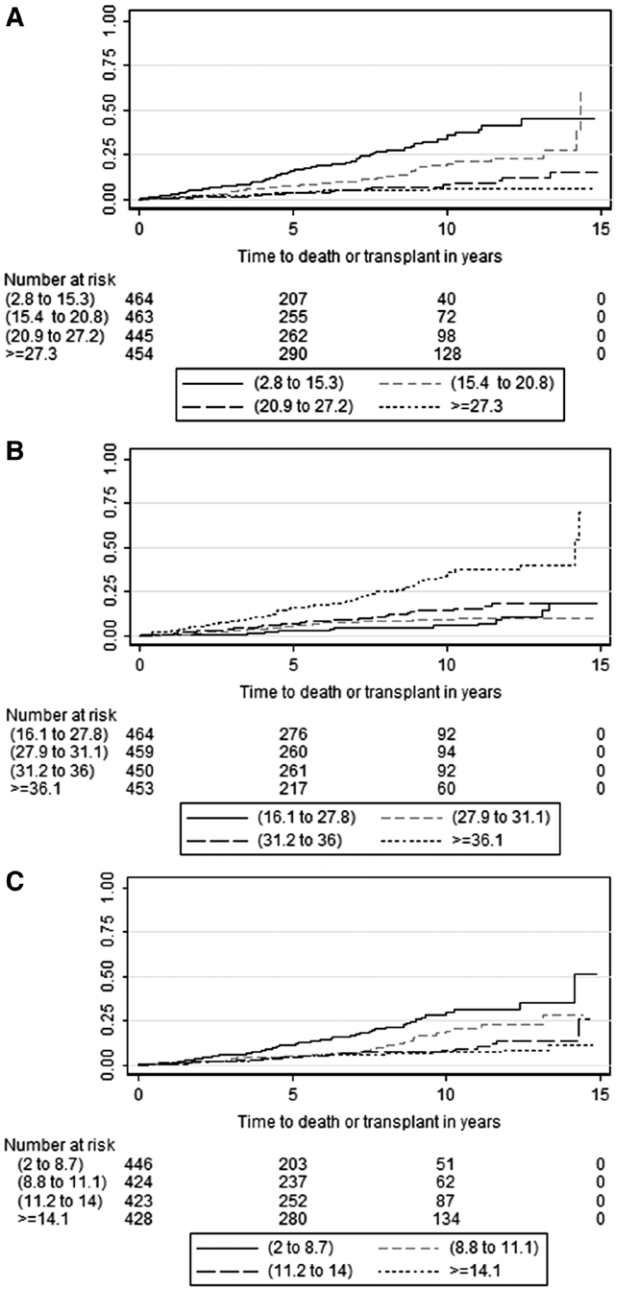


Figure 1. Kaplan–Meier analysis showing cumulative hazard rates of death/transplant in patients with hypertrophic cardiomyopathy (HCM) stratified according to key exercise parameters. **A**, Kaplan–Meier analysis stratified according to quartiles of peak oxygen consumption (peak $\dot{V}O_2$). Log rank <0.001; χ^2=98.05. **B**, Kaplan–Meier analysis stratified according to quartiles of ventilatory slope ($\dot{V}_ECO_2$). Log rank <0.001; χ^2=87.59. **C**, Kaplan–Meier analysis stratified according to quartiles of ventilatory anaerobic threshold (VAT). Log rank <0.001; χ^2=48.09.

Estimation of 5- and 10-Year Survival

The adjusted model using peak $\dot{V}O_2$ was used to estimate survival at 5 and 10 years. The overall adjusted death/transplant

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Table 5. Unadjusted Hazard Ratios (95% CI) for Secondary Outcomes (Cardiovascular Deaths, Heart Failure and Transplant, and Sudden Cardiac Death) Based on a Cox Regression Model

	Cardiovascular Deaths (Number of events=117)		Heart Failure/Transplant (Number of events=53)		Sudden Cardiac Death (Number of events=40)		SCD or Equivalent (Number of events=74)	
	HR (95% CI)	*P* Value	HR (95% CI)	*P* Value	HR (95% CI)	*P* Value	HR (95% CI)	*P* Value
CPX parameter								
Peak $\dot{V}O_2$, mL/kg/min	0.80 (0.75–0.86)	<0.001	0.81 (0.77–0.86)	<0.001	1.03 (1.00–1.06)	0.069	1.01 (0.98–1.03)	0.681
$\dot{V}_E\dot{V}CO_2$ Slope	1.07 (1.05–1.09)	<0.001	1.10 (1.08–1.13)	<0.001	1.01 (0.97–1.06)	0.549	1.00 (0.97–1.04)	0.847
VAT, mL/kg/min	0.77 (0.64–0.92)	0.004	0.77 (0.70–0.84)	<0.001	1.09 (1.01–1.16)	0.017	1.02 (0.97–1.08)	0.392

CI indicates confidence interval; CPX, cardiopulmonary exercise testing; HR, hazard ratio; VAT, ventilatory anaerobic threshold; and $\dot{V}O_2$, oxygen consumption.

estimates for patients within the lowest quartile (peak $\dot{V}O_2$ ≤15.3 mL/kg/min) were 14% at 5 years and 31% at 10 years. These estimates were close to the actual values observed in the population (Figure 3).

Discussion

This is the first study to examine the prognostic value of peak and submaximal exercise parameters in an unselected consecutive series of patients with obstructive and nonobstructive HCM. Independent of age and sex, peak $\dot{V}O_2$, $\dot{V}_E\dot{V}CO_2$ slope, and VAT predict survival.

Mechanisms of Exercise Intolerance in HCM

The mechanism of exercise limitation in patients with HCM is complex, but includes LVOTO, myocardial ischemia, and LV systolic and diastolic dysfunction. Previous studies in patients with HCM have shown that peak and submaximal metabolic exercise parameters are abnormal and that reduced stroke volume response, ventilation–perfusion mismatch, chronotropic incompetence, and abnormal peripheral oxygen utilization are potential contributors to exercise limitation.[37,38] Failure of stroke volume augmentation because of LV diastolic dysfunction has been confirmed with invasive hemodynamic studies, and a recent noninvasive study indicated that LVOTO exacerbates this mechanism, albeit with considerable interindividual variation.[39–42]

Gas Exchange Parameters and Survival

Numerous clinical features and measurements of cardiac function are associated with adverse outcomes in patients with heart disease, but CPX is unique in providing a single integrated assessment of central and peripheral hemodynamics, respiratory function, and skeletal muscle metabolism. Only one other study has examined the prognostic value of CPX in patients with HCM. In 182 minimally symptomatic patients with LVOTO, percentage of predicted peak $\dot{V}O_2$ was predictive of death and the onset of severe symptoms.[43] In this study, we demonstrated that peak $\dot{V}O_2$, VAT, and $\dot{V}_E\dot{V}CO_2$ slope predicted overall survival, but also found that when combined peak $\dot{V}O_2$ and $\dot{V}_E\dot{V}CO_2$ were independent predictors of HF and cardiac transplantation but not of SCD or its equivalent. This dissociation suggests that the mechanisms that predispose or cause sudden and fatal sustained ventricular arrhythmia are independent from the progressive loss of contractile function that occurs in many patients with HCM and illustrates the importance of prediction tools validated for specific end points.[31]

The $\dot{V}_E\dot{V}CO_2$ slope reflects ventilation–perfusion mismatch, as well as impairment of peripheral oxygen utilization. $\dot{V}_E\dot{V}CO_2$ has the advantage over peak $\dot{V}O_2$ of being linear, regardless of effort level, and it can be obtained reliably and reproducibly whether all the data points or only those after the VAT are taken into consideration. The mechanism of

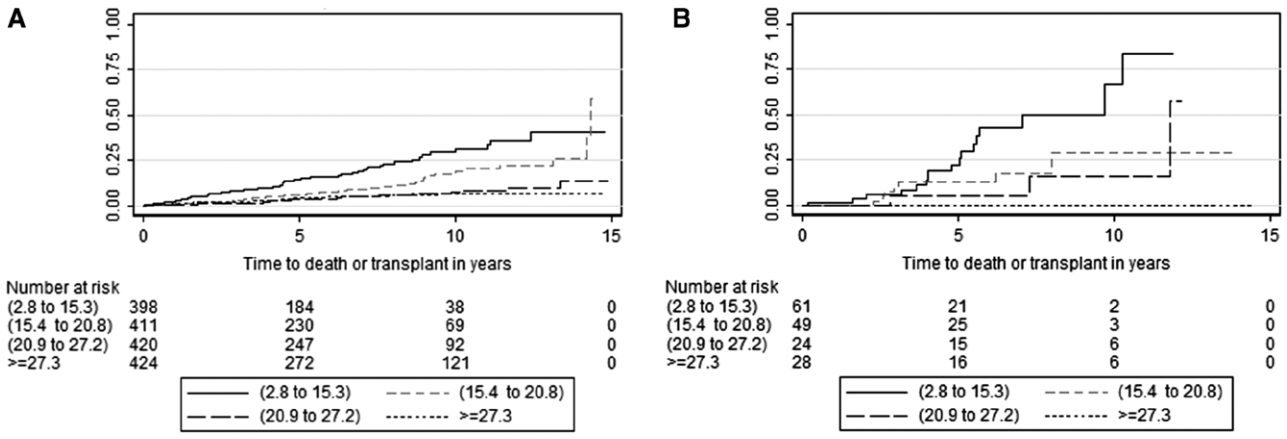


Figure 2. A, Kaplan–Meier analysis showing cumulative hazard rates of mortality/transplant in 1653 patients with hypertrophic cardiomyopathy (HCM) and ejection fraction (EF) ≥50% stratified according to quartiles of quartiles of peak oxygen consumption (peak $\dot{V}O_2$). Log rank <0.001; χ^2=72.3. **B**, Kaplan–Meier analysis showing cumulative hazard rates of mortality/transplant in 162 patients with HCM and EF <50% stratified according to quartiles of quartiles of peak oxygen consumption (peak $\dot{V}O_2$). Log rank <0.001; χ^2=20.1.

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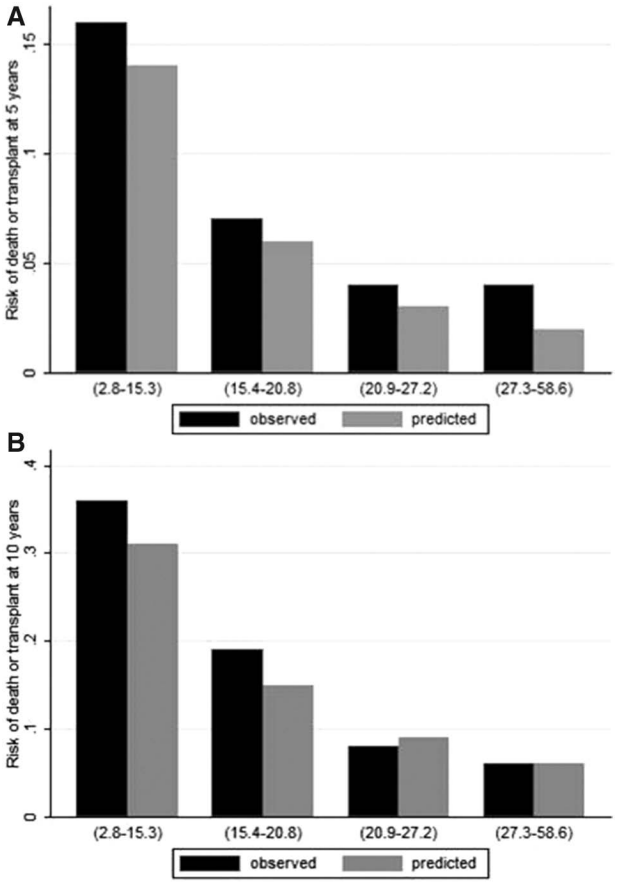


Figure 3. Observed and estimated survival based on the adjusted model using quartiles of peak $\dot{V}O_2$. **A**, 5-year survival. **B**, 10-year survival.

raised $V_E\dot{V}CO_2$ in HCM is difficult to determine in individual patients, but is likely to include reduced cardiac output, leading to ventilation perfusion mismatch, pulmonary microvascular dysfunction, and altered chemosensing because of resetting of the P_{CO_2} threshold and higher ventilation for lower P_{CO_2}, caused by repeated lactic acidosis at low effort. Although this study did not set out to determine the mechanism of elevated $V_E\dot{V}CO_2$ slope, we found higher values in patients with LVOTO.[38,44]

Exercise anaerobic threshold (VAT) could be determined in the majority of patients, but when combined with peak $\dot{V}O_2$ and $V_E\dot{V}CO_2$ slope, it was not a significant CPX predictor of overall survival. Patients who failed to reach anaerobic threshold could not be included in the regression model; however, they are an important cohort to identify and may have the worst prognosis. Their exclusion may explain why VAT does not perform as well as $V_E\dot{V}CO_2$ slope in this study as a predictor of transplantation or HF death. Previous investigators have shown VAT is interchangeable with peak $\dot{V}O_2$ in assessment of patients for cardiac transplantation ,and it may be useful in patients in whom motivation or other factors limit them from producing a true maximal effort.[19,45]

We found a worse overall prognosis in women who were also more likely to progress toward HF. This is consistent with emerging observational data from large cohorts and warrants further investigation.[46,47]

Clinical Implications

HCM is a complex disorder that follows different trajectories in individual patients. CPX aids in the diagnosis of secondary pathophysiologic consequences of HCM, for example, pulmonary hypertension, failure to augment stroke volume, and peripheral effects of detraining. It is also helpful diagnostically in differentiating hypertrophy because of athletic training, hypertension, or mitochondrial disease.[16,48]

Recent clinical guidelines recommend the use of CPX in baseline characterization and long-term follow-up of patients with HCM when the equipment and expertise are available.[49] These recommendations are based on consensus opinion and observational studies that have shown the utility of CPX in determining the severity and mechanism of exercise intolerance. The data in this study provide the first robust evidence to support these recommendations by showing that CPX parameters can also be used to identify patients at high risk of disease progression and early mortality from HF.

The design and implementation of pharmacological trials in HCM has been hampered by the heterogeneity of the disease and low event rates, but our study shows that it is possible to identify a cohort of patients who are at high risk for disease progression by measuring cardiorespiratory performance during exercise. Stratification of patients into high risk groups offers one way to increase the feasibility of trials, and the relation between CPX and outcomes suggests that the test may be a useful surrogate end point. This study shows that the prognostic utility of CPX applies equally to patients with and without resting LVOTO.

Limitations

This study was retrospective and performed at a tertiary center for HCM. Although the prevalence of exercise intolerance may be higher than in a community study, it is unlikely to affect the relation between exercise parameters and outcome, which was the main aim of the study. We did not systematically test for provokable obstruction, so it is not clear whether latent obstruction influenced CPX parameters. Although we evaluated the calibration and discrimination of the model used for the primary outcome, the generalizability of the model will depend on its performance in a new data set (external validity). This is an essential requirement for the model to be used routinely in clinical practice. Because of the study design, it was not possible to compare conventional exercise testing estimates of such as METS or exercise time with $\dot{V}O_2$. A direct comparison with standard exercise testing may be valuable. The association between exercise parameters and secondary outcomes was investigated using a similar approach to the primary outcome, but because of the problem of overfitting to the number of events, only the unadjusted analyses are presented.

Conclusions

This study provides evidence of the prognostic role of CPX testing in HCM and demonstrates that maximal and submaximal parameters are particularly useful in predicting HF-related outcomes.

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Disclosures

None.

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CLINICAL PERSPECTIVE

Exercise testing in patients with hypertrophic cardiomyopathy has traditionally been used to assess susceptibility to arrhythmia by measuring exercise blood pressure response and, in combination with echocardiography, to detect left ventricular outflow tract obstruction. Over the last decade, the value of cardiopulmonary exercise testing in predicting heart failure mortality has been confirmed by numerous observational studies, but a recent European Association for Cardiovascular Prevention and Rehabilitation/American Heart Association Joint Scientific Statement highlighted the need for additional data to support its use in diseases, such as hypertrophic cardiomyopathy. This large single-center study followed 1898 patients for median 5.6 years; 178 patients reached the primary end point of death or cardiac transplantation and demonstrated that peak oxygen consumption, ventilatory efficiency, and anaerobic threshold were independent predictors of the primary outcome. After correction for age, sex, left atrial size, nonsustained ventricular tachycardia, and left ventricular ejection fraction, the overall death/cardiac transplant estimates for patients with peak oxygen consumption ≤15.3 mL/kg/min were 14% at 5 years and 31% at 10 years. Importantly, peak oxygen consumption and ventilatory efficiency were predictors of death because of heart failure or transplantation but not sudden cardiac death or implantable cardioverter defibrillator shocks. This study demonstrates that cardiopulmonary exercise testing provides important prognostic information in patients with hypertrophic cardiomyopathy. The findings also suggest that cardiopulmonary exercise testing may provide a useful means for stratifying patients with hypertrophic cardiomyopathy to power clinical trials of existing and novel therapies targeting hard survival end points.